

Lu YU

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Education

- 2025–present **Postdoc in Applied Mathematics**, *ENPC ParisTech*, Paris, France
- 2022–2025 **Ph.D. in Applied Mathematics**, *University Paris 1*, Paris, France
Dissertation: Theory of Fixed Points with Discontinuities and Applications in Game Theory
Advisor: Philippe Bich
- 2021–2022 **Master in Modeling and Mathematical Methods in Economics and Finance**,
University Paris 1, Paris, France
Très bien
Master thesis: Fixed-Point Theorems for Discontinuous Mappings on Topological Vector Space.
- 2019–2021 **Master in Fundamentals Mathematics**, *University Paris 7*, Paris, France
Master thesis: Asymptotic of the Bergman kernel on a projective complex manifold.
- 2013–2017 **Bachelor in Mathematics**, *Ocean University of China*, QingDao, China

Publications

- Lu Yu. Fixed point theorems for increasing correspondences on lattices. *Economic Theory*, 2026
- Xin Lai, Shiming Deng, Lu Yu, Yumin Lai, Shenghao Qiao, and Xinze Zhang. Rethinking Recurrent Neural Networks for Time Series Forecasting: A Reinforced Recurrent Encoder with Prediction-Oriented Proximal Policy Optimization. *Advanced Engineering Informatics*, Vol. 76, 2026
- Lu Yu. The degree sequence of the preferential attachment model. *Discrete Applied Mathematics*, Vol. 363, pp. 158-167, 2025.
- Lu Yu. Nash equilibria of quasisupermodular games. *Operations Research Letters*, Vol. 58, 2025.
- Lu Yu. Remarks on “some fixed point theorems for s -convex subsets in p -normed spaces”. *The Journal of Analysis*, Vol. 32, No. 2, pp. 1139-1143, 2023.
- Lu Yu. Existence and structure of Nash equilibria for supermodular games. *Order*, pp.1-12, 2024.
- Lu Yu. Notions of relative interior for compact convex sets. to appear in *Journal of Convex Analysis*, 2024.
- Lu Yu. Generalization of Zhou fixed point theorem. *International Game Theory Review*, 2024.

Preprints under review

- Lu Yu. Nash equilibria of games with generalized complementarities. minor revision

Preprints

- Philippe Bich and Lu Yu. A Brouwer-Tarski fixed-point theorem.

Lu Yu. Monotone comparative statics of parameterized games.
 Lu Yu. Nash equilibria of games with generalized complementarities.
 Lu Yu. A degree theoretical proof of Singbal's fixed point theorem. submitted.
 Lu Yu. Nash equilibria in discontinuous and non-quasiconcave games.
 Lu Yu. Existence of pure stationary equilibria for stochastic games.
 Lu Yu. Discounted stochastic game with unbounded costs on compact action spaces.

Talks

Conference

- 2024.10.28-10.30 Young Researchers in Applied Mathematics conference, ENS Lyon, Lyon, France
 2025.01.06-01.10 Asian School in Economic Theory, Abu Dhabi, the United Arab Emirates
 2026.02.24-02.26 Congrès ROADEF 2025, Tours, France

Seminar

- 2026.01. Paris Game Theory Seminar, Institut Henri Poincaré, Paris, France
 2023.06.23 Mathematics Day in University Paris 1, Paris, France
 2024.10.14 Paris Junior Game Theory Seminar, Institut Henri Poincaré, Paris, France
 2026.01.26 Paris Game Theory Seminar, Institut Henri Poincaré, Paris, France
 2026 CERMICS Student Seminar, CERMICS, École des Ponts ParisTech, Champs-sur-Marne, France

Workshop

- 2024.07.10-07.12 Sorbonne Workshop in Economic Theory (poster session), Venice, Italy
 2024.12.05-12.07 IX Hurwicz Workshop on Mechanism Design Theory(Chair for session Mathematical Economics), Warsaw School of Economics, Banach Center, Warsaw, Poland
 2026.06.28-07.12 2026 Workshop on Topological Combinatorics, GIST, Gwangju, Republic of Korea

Teaching experience

- 2022 S1 **Mathematics foundation**, *Teaching assistant for exercise session 24h, 30 students*, University Paris 1
 2022 S2 **Analysis mathematics**, *Teaching assistant for exercise session 24h, 30 students*, University Paris 1
 2023 S2 **Analysis mathematics**, *Teaching assistant for exercise session 48h, 60 students*, University Paris 1
 2026 Supervised 4 undergraduate (École des Ponts ParisTech) Bachelor's theses. topic: Sur l'intérieur relatif des ensembles convexes compacts dans les espaces vectoriels topologiques.

Skills

Programming Python, Matlab, L^AT_EX

Languages English, French, Chinese

Miscellaneous

I am a chemistry enthusiast and I have previously worked as a chemical analyst at Qitai Petrochemical